

Lab 4: Lines and Planes in $\mathbb{R}^3$

Math 124, Fall 2026 at the University of Michigan

due by the end of class on Wednesday, September 23, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, most of which will involve writing math on paper, and some of which will involve running code in a Jupyter Notebook. To receive credit for today's lab, you must show your lab TA your work on this worksheet.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Today's lab is split into two parts, all of which are covered in Chapter 2.4 of the course notes:

- Activities 1-4 involve content from the previous two lectures.
- Activities 5-8 involve affine lines and planes in $\mathbb{R}^3$ – that is, objects that don't necessarily pass through the origin, $(0, 0, 0)$. We will give a brief overview of these in lab and revisit them in tomorrow's lecture. (Activities 5-8 are the same as Questions 5-8 from yesterday's lecture worksheet, which we didn't get to.)

Activity 1: The Span of One Vector

The line ℓ is given in scalar-parametric form by

$$x = 5t, \quad y = -9t, \quad z = 2t, \quad t \in \mathbb{R}.$$

- a) Find a vector $\vec{u}$ such that $\ell = \text{span}(\vec{u})$.

- b) Find a different vector with the same span. Explain why your vector describes the same line.

Activity 2: The Same Plane, Different Directions

The plane P is given in scalar-parametric form by

$$x = s + t, \quad y = s - t, \quad z = 2t, \quad s, t \in \mathbb{R}.$$

- a) Find vectors $\vec{u}$ and $\vec{v}$ such that $P = \text{span}(\vec{u}, \vec{v})$.

- b) Are $\vec{u}$ and $\vec{v}$ the only pair of vectors that span P ? Compute $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$, and explain why this pair also spans P . *Hint: Can you write each of $\vec{u}$ and $\vec{v}$ as a linear combination of $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$?*

- c) Use $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ to write a different scalar-parametric form for P . Write your answer as equations for x , y , and z , and state the possible values of your parameters.

Activity 3: A Plane in Two Forms

Let

$$\vec{u} = \begin{bmatrix} 5 \\ -7 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}, \quad P = \text{span}(\vec{u}, \vec{v}).$$

- a) Write P in scalar-parametric form by writing three separate equations: one for x , one for y , and one for z . All three should use the same two parameters, $s, t \in \mathbb{R}$.

- b) Find a nonzero vector $\vec{n}$ that is perpendicular to both $\vec{u}$ and $\vec{v}$. Verify your answer using dot products.

- c) Use $\vec{n}$ to write a linear equation of the form $ax + by + cz = 0$ for P . Graph your equation on Desmos, [desmos.com/3d](https://www.desmos.com/3d).

Activity 4: A Line in Equation Form

Consider the line from Activity 1:

$$\ell = \text{span} \left(\begin{bmatrix} 5 \\ -9 \\ 2 \end{bmatrix} \right).$$

- a) Find two vectors that are both orthogonal to $\begin{bmatrix} 5 \\ -9 \\ 2 \end{bmatrix}$ and are not scalar multiples of each other, i.e. two linearly independent vectors on $\ell^\perp$.

- b) Use those vectors to write a system of two linear equations and three unknowns (x , y , and z) whose solution set is the line ℓ .

- c) Graph both equations on Desmos, desmos.com/3d. What do they look like?

The remaining activities are about affine lines and planes – that is, objects that don't necessarily pass through the origin, $(0, 0, 0)$. These are discussed at the bottom of Chapter 2.4. We will give a brief overview ~an hour into lab to help you get started.

Activity 5: An Affine Line in Parametric Form

Let $\ell' = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

- a) Write ℓ' explicitly in vector-parametric form.

$$\ell' = \left\{ \begin{bmatrix} \\ \\ \end{bmatrix} + t \begin{bmatrix} \\ \\ \end{bmatrix} \mid t \in \mathbb{R} \right\}.$$

- b) Write the corresponding scalar-parametric equations for x , y , and z .

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}},$$

$$z = \underline{\hspace{2cm}}.$$

- c) Graph ℓ' on Desmos, [desmos.com/3d](https://www.desmos.com/3d). To get the entire line to appear, you may have to change the upper and lower bounds for t .

Activity 6: An Affine Plane in Parametric Form

Now let's switch to discussing affine planes. Let $P' = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

- a) Write P' explicitly in vector-parametric form using two parameters s and t .

$$P' = \left\{ \begin{bmatrix} \\ \\ \end{bmatrix} + s \begin{bmatrix} \\ \\ \end{bmatrix} + t \begin{bmatrix} \\ \\ \end{bmatrix} \mid s, t \in \mathbb{R} \right\}.$$

- b) Write the corresponding scalar-parametric equations for x , y , and z .

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}},$$

$$z = \underline{\hspace{2cm}}.$$

- c) Graph P' on Desmos. Start with <https://www.desmos.com/3d/llepnyv69l> (found by Googling "plane in parametric form on Desmos") and make modifications as necessary.

Activity 7: An Equation for an Affine Plane

Consider the affine plane from Activity 6: $P' = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

- a) Find a nonzero normal vector $\vec{n}$ perpendicular to both spanning vectors above.

- b) A point $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ lies in P' when $\vec{n} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n} \cdot \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

Compute the constant on the right-hand side.

$$\vec{n} \cdot \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = \underline{\hspace{2cm}}.$$

- c) Write an equation for P' in dot-product form.

$$\vec{n} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underline{\hspace{2cm}}.$$

- d) Write a linear equation of the form $ax + by + cz = d$ for P' .

- e) Graph your equation for P' on Desmos, [desmos.com/3d](https://www.desmos.com/3d). Compare the graph with the one you made in Activity 6(c).

Activity 8: An Affine Line as the Solution of a Linear System

Finally, let's return to the affine line from Activity 5: $\ell' = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \text{span} \left(\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \right)$.

- a) Find two normal vectors $\vec{n}_1$ and $\vec{n}_2$ perpendicular to the direction vector above that are not scalar multiples of each other.

- b) A point $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ lies on ℓ' when $\vec{n}_1 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n}_1 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $\vec{n}_2 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{n}_2 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

Compute the two constants on the right-hand sides.

$$\vec{n}_1 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \underline{\hspace{2cm}}, \quad \vec{n}_2 \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \underline{\hspace{2cm}}.$$

- c) Write a system of two linear equations and three unknowns (x , y , and z) whose solution set is ℓ' .

- d) Graph both equations in your system on Desmos, desmos.com/3d. The two planes should intersect in the affine line ℓ' . Compare the intersection with the graph from Activity 5(c).